

Registration No.:

Course Code: CSE306
Course Title: COMPUTER NETWORKS

23241CSE22154
Paper Code: A

Max Marks: 70

Time Allowed: 03:00hrs.

- Read the following instructions carefully before attempting the question paper.
1. Match the Paper Code shaded on the OMR Sheet with the Paper code mentioned on the question paper and ensure that both are the same.
 2. This question paper is divided into two parts A and B.
 3. Part A contains 30 questions of 1 mark each. 0.25 marks will be deducted for each wrong answer.
 4. Part B contains 5 questions of 10 marks each. Attempt any 4 questions out of these 5 questions. In case all the 5 questions are attempted then only the first four attempted questions will be evaluated.
 5. Attempt all the questions in serial order.
 6. Do not write or mark anything on the question paper except your registration no. on the designated space.
 7. After completion of first 90 minutes, the OMR sheet will be taken by the invigilator.
 8. Submit the question paper and the rough sheet(s) along with the answer sheet to the invigilator before leaving the examination hall.

Part-A

- Q1) _____ is a high performance fiber optic token ring LAN running at 100 Mbps over distances upto 1000 stations
(a) FDDI (b) FDDT (c) FDDR (d) FOTR CO1,L3
- 2) In distance vector routing the dynamic updates are shared
(a) Periodically (b) When there is a topology change (c) Both (A) and (B) are correct (d) None of these CO1,L3
- 3) A _____ is a TCP name for a transport service access point.
(a) port (b) pipe (c) node (d) protocol CO5,L1
- 4) Which is not a type of extension header:
(a) source routing (b) authentication (c) encrypted option (d) none of above CO6,L4
- 5) The higher layer of SSL includes a mechanism for initial authentication and key transmission known as the
(a) SSL handshake protocol (b) SSL authentication protocol (c) SSL record protocol (d) SSL cipher protocol CO1,L3
- 6) Which of the following are types of PORT?
(a) Well Known (b) Registered (c) Dynamic (d) All of the above CO6,L4
- 7) When a web browser needs to resolve the IP address of "www.example.com," which DNS server does it typically query first?
(a) The authoritative DNS server for example.com (b) The root DNS server (c) The local DNS server configured on the user's device (d) The ISP's DNS server CO1,L4
- 8) In classless addressing, there are no classes but the addresses are still granted in
(a) Codes (b) Sections (c) Codewords (d) Blocks CO4,L3
- 9) During the TCP three-way handshake, which step involves the client requesting a connection by sending
(a) SYN (b) SYN-ACK (c) ACK (d) FIN CO1,L3
- 10) A leaky bucket algorithm shapes bursty traffic into fixed-rate traffic by averaging _____
(a) Data Rate (b) Flow Rate (c) Error Rate (d) Access Rate CO5,L4
- 11) In distance vector routing, a router has a route as Destination:N, Hop Count:7, Next Hop:B. It receives a packet from C with a route to N as Destination: N, as Destination :N, Hop Count :7. the new route to N should be
(a) Destination: N, Hop Count: 7, Next Hop:B (b) Destination: N, Hop Count: 7, Next Hop:C (c) Destination: N, Hop Count: 6, Next Hop:C (d) none of the above CO5,L4
- 12) Apply your knowledge of IPv4 header and tell which statement is true about the IPv4 header?
(a) Only the header in the datagram is fragmented (b) Only the data in the datagram is fragmented (c) Both header and data in the datagram are fragmented (d) None of the above CO4,L3
- 13) Statement I: These Domain name starts from a node, but it does not reach the root
Statement II: Example is myserver.com
The above two statements define which of the following Domain Name?
(a) Zone Qualified Domain Name (b) Partially Qualified Domain Name (c) Country Qualified Domain Name (d) Fully Qualified Domain Name CO5,L4

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- 14) Indicate starvation is caused in which queue?
(a) FIFO (b) Priority queue (c) LIFO (d) Stack
- 15) Each block in class A contains _____ addresses.
(a) 2^{16} (b) 2^{24} (c) 2^8 (d) 2
- 16) Analyze in case of transport layer, processes communicate with each other by which of the addresses?
(a) Specific Addresses (b) Mac Addresses (c) Port Addresses (d) None of the above
- 17) Compute if the fragmentation offset value is 200, then what will be the value of the first byte?
(a) 3200 (b) 1200 (c) 1600 (d) 5100
- 18) _____ is used to record time of datagram processing by a router.
(a) Timestamp (b) Time frame (c) Time wrap (d) Time delay
- 19) _____ have problem of starvation:
(a) FIFO queuing (b) Priority queuing (c) Weighted Fair queuing (d) none of above
- 20) Examine how many bits are used for the network portion in a Class B IP address?
(a) 8 bits (b) 16 bits (c) 24 bits (d) 32 bits
- 21) In segment header, sequence number and acknowledgement number fields refer to _____
(a) Byte number (b) Buffer number (c) Segment number (d) Acknowledgment
- 22) What is the First Usable IP on the subnetwork that the node 192.168.1.15/28 belongs to?
(a) 192.167.1.1 (b) 192.168.1.1 (c) 192.168.0.1 (d) 192.167.0.1
- 23) What is the standard form of ICMP?
(a) Internet Control Message Particle (b) Internet Control Message Point (c) Internet Control Message Protocol (d) None of the above
- 24) The normal size of the TCP header is how many bytes?
(a) 24 bytes (b) 28 bytes (c) 16 bytes (d) 20 bytes
- 25) Which of the following is not a type of Closed-loop Congestion Control?
(a) Back Pressure (b) Window Policy (c) Choke Packet (d) Implicit Signaling
- 26) Closed-Loop control mechanisms try to _____
(a) Remove after sometime (b) Prevent before congestion occurs (c) Remove after congestion occurs (d) None of these
- 27) Illustrate the size of Source and Destination IP address in IP header?
(a) It means addressing a packet to all machine (b) It means addressing a packet to some machine (c) It means addressing a packet to a particular machine (d) It means addressing a packet to except a particular machine
- 28) Transmission control protocol _____
(a) is a connection-oriented protocol (b) uses a three way handshake to establish a connection (c) receives data from application as a single stream (d) all of the mentioned
- 29) Identify the maximum number of unique ip addresses that can be assigned within a single subnet of 255.255.255.0 (CIDR/24)
(a) 16 (b) 256 (c) 512 (d) 1024
- 30) Retransmission of packets must not be done when
(a) Packet is needed (b) Packet is error-free (c) Packet is lost (d) none of above

Part-B

- Q2). 1. What are the various causes of transmission impairment while transmitting analog signals?
2. How does the process of digital to analog conversion shape our perception of the analog world, and what implications does it hold for the human experience in an increasingly digital age?
- Q3). Compare and contrast connectionless and connection oriented service with the help of example
- Q4). Write a short note on routing. Explain shortest path algorithm and distance vector routing with suitable examples.
- Q5). Explain the difference between classful and classless ip addressing scheme and which one is the most effective method of allocating IP address?
- Q6). Explain the concept of random access protocols of data link layer with the help of suitable scenario's?

-End of Question paper-